

DeepThinkTrader v3.0 — Complete System Guide

Generated: March 24, 2026 | **Repository:** github.com/RebelHawk-TK/DeepThinkTrader

CRITICAL WARNING: This is a paper trading bot. Never use real money until backtested for 3+ months. Trading bots can lose 100% of capital. This project is educational and experimental.

1. Architecture Overview

Pipeline: Scanner (RSI-scored) → Research Agent (+ Obsidian SA) → DeepThink Analysis (Rule-Based + Multi-Edge + Edge Learning) → Risk Gate (15 checks) → Execution Agent (spread/VIX/sector/gap guards) → Alpaca Paper Trading

Component	Technology	Purpose
Scanner Agent	Alpaca Screener API + RSI	3-stage funnel with RSI momentum scoring — main + penny (\$1-\$5)
Research Agent	Alpaca + Twelve Data + NewsAPI + Yahoo + Obsidian SA	Gathers technicals, fundamentals, news, Reddit, Seeking Alpha email intelligence
DeepThink Agent	Rule-based + edge learning + sector rotation	Conviction scoring with historical edge combo win rates, sector trend awareness
Risk Manager	Kelly criterion + 15 portfolio guards	Spread, VIX, sector concentration, gap risk, earnings hours, plus original 13 checks
Execution Agent	Alpaca Paper Trading API	Market/limit orders, order polling, partial fill retry, slippage analytics, momentum exits
Dashboard	Streamlit + Plotly	Real-time portfolio monitoring, trade history, strategy health
Database	SQLite	Trades, ATR history, slippage records, edge performance, partial exits, request IDs

2. Data Sources

Source	Data Provided	API Key Required	Free Tier
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Alpaca Markets	Price, volume, SMA, RSI, trade execution, portfolio history, SPY snapshots	Yes (paper account)	Unlimited
Twelve Data (RapidAPI)	MACD, Bollinger Bands, EMA 9/21, Stochastic, ADX, ATR	Yes (RapidAPI key)	800 req/day
NewsAPI	News headlines + VADER sentiment scoring	Yes	500 req/day
Yahoo Finance	P/E, ROE, growth, debt/equity, earnings dates, insider trades, institutional holdings	No	Unlimited
Reddit (PRAW)	Sentiment from r/wallstreetbets, r/stocks, r/investing, r/pennystocks	Yes (optional)	60 req/min
Obsidian Vault	Seeking Alpha newsletter emails — ticker mentions, sentiment, categories	No (local files)	N/A

3. Risk-First Framework (v3.0)

Philosophy: Every trade must pass through a rigorous, emotion-proof checklist before execution. No trade executes on conviction alone — it requires multi-edge confirmation, portfolio-level safety, and execution quality validation (spread, VIX, sector, gap risk).

3.1 Position Sizing: Fractional Kelly

Condition	Method	Formula
20+ closed trades	Fractional Kelly	$f^* = (p - q) / b \times 0.5$ (half-Kelly)
< 20 trades	Fixed Risk	1% of equity per trade

Where p = win rate, $q = 1 - p$, b = avg win / avg loss (payoff ratio). Capped at MAX_POSITION_PCT.

3.2 Pre-Trade Checks (16 gates)

#	Check	Action if Failed
0	Warmup complete (200+ unique tickers analyzed)	BLOCKED (all trades until warmup done)
1	Conviction meets threshold	HOLD
2	Risk within limit	BLOCKED
3	Reward:Risk ratio acceptable	BLOCKED
4	Daily loss limit OK	BLOCKED for rest of day
5	Open position count OK	BLOCKED
6	No duplicate position	BLOCKED
7	Market hours	BLOCKED
8	Drawdown from peak < 8%	BLOCKED (all new entries)
9	Risk of ruin < 1%	BLOCKED
10	Liquidity: shares < ADV/5	Auto-reduce or BLOCKED
11	Multi-edge: 2/3 edges firing	HOLD
12	Market health: SPY down > 2% OR VIX ≥ 30	BLOCKED (SPY=longs, VIX=all)
13	No earnings within 2 days / < 12 hours	Auto-close or BLOCKED

14	Bid-ask spread acceptable (< 1% main, < 2% penny)	BLOCKED
15	Sector concentration < 25% of portfolio	BLOCKED

3.3 Volatility Adjustment

If current ATR > 3× median ATR (real 30-day median from stored history), risk is cut in half.

3.4 Overnight Gap Risk (v3.0)

Position size is automatically reduced for high-gap-risk names based on ATR/price %, beta, and earnings proximity. High risk (score ≥ 2) = 50% reduction, moderate (score ≥ 1) = 75%.

3.4 Revenge Trading Detector

If last 3 trades are all losses, all new entries are blocked until a winning trade occurs.

4. Multi-Edge Validation (v2.0)

Rule: A trade requires at least **2 of 3 independent edges** to execute. Conviction score alone is not enough.

4.1 Fundamental Edge

Must pass at least 3 of 4 criteria:

Criterion	Threshold
P/E ratio	< 20 or forward P/E improving
Return on Equity	> 15%
Earnings/Revenue Growth	> 10% YoY
Debt/Equity	< 100

4.2 Technical Edge

Must pass at least 2 of 3 criteria:

Criterion	Threshold
Price vs 200-day SMA	Above (for longs)
RSI(14)	< 45 (neutral-bullish) or < 30 (oversold)
Volume breakout	> 1.5× 20-day average

4.3 Sentiment/Regime Edge

Must pass at least 2 of 3 criteria:

Criterion	Threshold
Combined catalyst score	> 0.2
News sentiment	> 2/10 (bullish)
Reddit sentiment	> 0.3 (bullish)

5. Exit Management (v2.0)

5.1 Exit Check Frequency

Check Type	Frequency	What It Does
Fast exit check	Every 5 minutes	Price checks only — SL/TP/trailing/time stops
Full cycle	Every 60 minutes	Full scan + research + analysis + execution

5.2 Trailing Stops

Parameter	Main Portfolio	Penny Portfolio
Activation	2% profit	2% profit
Trail distance	1.5%	3.0%

Once activated, trailing stop replaces static stop-loss. Static take-profit still applies.

5.3 Partial Scale-Out

Profit Level	Action	Remaining
1R (1× risk amount)	Sell 33% of position	67%
2R (2× risk amount)	Sell another 33%	34%
3R+ or trailing stop	Remaining 34% rides with trail	0%

5.4 Time Stop

If a position shows < 2% movement after 15 trading days, it is automatically closed to free capital.

5.5 Earnings Auto-Exit (v3.0)

If earnings are within 2 trading days OR < 12 hours away (imminent flag), position is automatically closed. Pre-market earnings at 7am are now caught even if checked at 3pm the day before.

5.6 Momentum Divergence Exit (v3.0)

On every exit check, profitable positions are checked for RSI divergence: RSI > 75 on a long = tighten stop to 50% distance. Weak volume + overbought = same tightening. This locks in profits before momentum fades.

6. Smart Order Execution (v2.0)

6.1 Order Types

Portfolio	Order Type	Details
Main	Market order	Immediate fill, slippage tracked
Penny	Limit order	Current price + 0.5% slippage buffer, day TIF, auto-cancel after 30 min

6.2 Slippage Tracking

After fill, actual price is compared to expected. Deviations > 0.3% are logged as slippage alerts.

6.3 Pre-Trade Transparency

Every trade logs a plain-English summary before execution:

```
BUY 120 shares of XYZ @ $4.50
Risk: $54 (0.1% of portfolio) | Edges: 2/3
R:R = 2.4:1 | Stop: $4.05 | Target: $5.58
Portfolio: penny | ADV: 500,000
```

7. Market Circuit Breaker (v3.0)

Trigger	Action
SPY down > 2% intraday	Block all new LONG entries (shorts still OK)
VIX ≥ 30	Block ALL new entries (extreme fear regime)
30-day drawdown > 8%	Block ALL new entries (not exits)
Negative expectancy	Block trades (risk of ruin too high)

8. Post-Trade Learning & Intelligence (v3.0)

8.1 Strategy Health Metrics

Metric	Description
Win Rate (30d)	% of trades that were profitable
Expectancy	(win_rate × avg_win) - (loss_rate × avg_loss)

Profit Factor	Gross wins / gross losses
Payoff Ratio	Avg win / avg loss
WR vs Baseline	30-day win rate vs 90-day baseline

8.2 Kelly Feedback Loop

Win rate and payoff ratio from closed trades feed directly into position sizing. As strategy performs better, Kelly sizes up. As it degrades, sizing shrinks automatically.

8.3 Weekly Health Check (Monday)

If 30-day win rate drops > 15% from 90-day baseline, a degradation warning is logged.

8.4 Slippage Analytics (v3.0)

Every fill is recorded with expected vs actual price, order type, and hour of day. Aggregated by ticker and time to identify costly patterns. Before trading, the system warns if a ticker historically has > 0.5% average slippage.

8.5 Edge Combo Learning (v3.0)

When trades close, the edge combination used (F=Fundamental, T=Technical, S=Sentiment) is recorded with the P&L outcome. Over time, combos with > 65% win rate get +1.5 conviction boost, while combos < 35% get -1.5 penalty. This creates a self-improving feedback loop.

8.6 Sector Rotation Awareness (v3.0)

Before entry, the system checks if the stock's GICS sector ETF (XLK, XLV, XLF, etc.) is above its 50-day SMA. Weak sectors get a -0.5 conviction penalty and a bearish factor. This prevents buying strong stocks in weak sectors.

8.7 Obsidian / Seeking Alpha Integration (v3.0)

The system reads Seeking Alpha newsletter emails from an Obsidian vault, extracting:

Data Point	How Used
Ticker mentions	Boost catalyst score for mentioned tickers
Sentiment	VADER scoring on context around each ticker
Categories (earnings, insider, analyst, etc.)	Feed into risk/opportunity signals

9. Configuration

9.1 Trade Modes

Parameter	Safe	Normal	Aggressive
Max Risk/Trade	1%	2%	3%
Daily Loss Limit	3%	5%	8%
Min Conviction	9.0	7.5	6.0
Min R:R Ratio	3:1	2:1	1.5:1
Max Position %	5%	10%	15%
Max Positions	5	10	15

9.2 Risk-First Parameters (v2.0)

```
# Risk Gate
RISK_PCT_PER_TRADE=0.01      # 1% default (Kelly fallback)
MAX_DRAWDOWN_HALT_PCT=0.08   # 8% drawdown halts trading
VOLATILITY_ATR_MULTIPLIER=3.0 # ATR > 3x median = cut risk 50%
MIN_ADV_RATIO=5              # Shares must be < ADV/5
KELLY_SAFETY_MULTIPLIER=0.5   # Half-Kelly
MAX_RISK_OF_RUIN_PCT=0.01     # Block if RoR > 1%

# Exit Management
EXIT_CHECK_INTERVAL_MINUTES=5
TRAILING_STOP_ACTIVATION_PCT=2.0
TRAILING_STOP_DISTANCE_PCT=1.5
PENNY_TRAILING_STOP_DISTANCE_PCT=3.0
SCALE_OUT_ENABLED=true
SCALE_OUT_LEVELS=1.0,2.0      # R-multiples
TIME_STOP_DAYS=15

# Edge Validation
MIN_EDGES_REQUIRED=2          # 2 of 3 edges must fire

# Smart Orders
PENNY_LIMIT_SLIPPAGE_PCT=0.5
MAX_SLIPPAGE_PCT=0.3

# Market Awareness
CIRCUIT_BREAKER_SPY_DROP_PCT=-2.0
CIRCUIT_BREAKER_VIX_THRESHOLD=30.0
EARNINGS_EXIT_DAYS=2
```

```
EARNINGS_EXIT_MODE=close          # or "tighten"

# Execution Quality (v3.0)
MAX_SPREAD_PCT=1.0                 # Block if bid-ask spread > 1%
PENNY_MAX_SPREAD_PCT=2.0
MAX_SECTOR_EXPOSURE_PCT=0.25      # 25% max per GICS sector
GAP_RISK_ATR_THRESHOLD=5.0        # ATR% > 5 = high gap risk
GAP_RISK_POSITION_REDUCTION=0.5   # Cut position 50% for high gap risk

# Obsidian Integration (v3.0)
OBSIDIAN_VAULT_PATH=~/Documents/RHVault/RHVault
OBSIDIAN_SA_MAX_AGE_DAYS=7

# Claude AI Analysis (v3.0)
CLAUDE_ANALYSIS_ENABLED=true       # Enable/disable LLM analysis
CLAUDE_MODEL=claude-haiku-4-5-20251001 # Model for per-ticker calls
```

10. Claude AI Analysis Layer (v3.0)

Optional: When an Anthropic API key is configured, Claude provides qualitative judgment on every trade setup — catching what rule-based scoring misses.

Capability	What It Catches
News Interpretation	Sarcastic headlines, misleading sentiment, one-time events that VADER misscores
Signal Independence	When "3 edges firing" are actually correlated (all driven by same catalyst)
Earnings Quality	Sustainable growth vs accounting tricks, one-time beats
Risk Synthesis	Connecting multiple risk factors into a coherent narrative
Contrarian Reasoning	Qualitative pushback beyond template contrarian views

10.1 How It Works

Claude receives the full research report + rule-based analysis, then returns:

Field	Range	Effect
conviction_adjustment	-2.0 to +2.0	Scaled by confidence, added to rule-based conviction
action_override	BUY/SELL/HOLD/null	Only applied at 80%+ confidence
confidence	0 to 1	Scales the conviction adjustment (low confidence = small impact)

10.2 Cost

Uses Claude Haiku by default (~\$0.001 per ticker analysis). At 20 tickers per cycle, that's ~\$0.02/hour or ~\$0.30/day during market hours.

11. Security Hardening (v3.0)

Area	Protection
API Keys	macOS Keychain (not in code or .env)
Database	File permissions 600 (owner-only), set on creation
Log Files	Permissions 600, rotating (10MB max, 5 backups)

PID Files	Permissions 600, created atomically
Obsidian Vault	Path traversal blocked via realpath() validation
YAML Parsing	safe_load() only, 1MB file size limit
SQL Queries	Parameterized throughout (no string interpolation)
Network	HTTPS on all external APIs, TLS cert validation
.env File	Gitignored, never committed

12. Quick Start

```
git clone https://github.com/RebelHawk-TK/DeepThinkTrader.git
cd DeepThinkTrader
cp .env.template .env          # Fill in your API keys
pip install -r requirements.txt
python main.py once             # Test single cycle
python main.py scan             # Test scanner
python main.py penny scan      # Test penny scanner
python main.py ticker NVDA     # Test single ticker with edge validation
streamlit run dashboard.py     # Launch dashboard
./run.sh                       # Start bot + dashboard
./stop.sh                      # Stop everything
```

13. Project Structure

```
StockTrader/
├─ main.py                # Orchestrator (log rotation + scheduled loop)
├─ config.py              # Environment config (3 modes + 25 risk params)
├─ dashboard.py           # Streamlit (ticker bar + benchmarks + health)
├─ run.sh / stop.sh / status.sh # Control scripts (secure PID/log perms)
├─ install.sh             # macOS installer (venv + apps)
├─ DeepThinkTrader.app/   # macOS app – double-click to start
├─ DeepThinkTrader Stop.app/ # macOS app – double-click to stop
├─ agents/
│   ├─ deepthink_agent.py # Rule-based + Claude AI + edge learning
│   ├─ execution_agent.py # Alpaca trades + 15 safety checks
│   ├─ research_agent.py  # 7 data sources (Alpaca/12Data/News/Reddit/Yahoo/SA RSS/Ob
│   └─ scanner_agent.py   # RSI-scored 3-stage discovery (main + penny)
├─ utils/
│   ├─ alpaca_data.py     # Alpaca market data + X-Request-ID
│   ├─ claude_analyst.py  # Claude AI qualitative analysis layer
│   ├─ database.py        # SQLite (9 tables, 600 perms, parameterized queries)
│   └─ market_clock.py    # Alpaca market clock API + timezone handling
```

```
| |— obsidian_seeking_alpha.py # SA email parser (path traversal protected)
| |— risk_manager.py          # Kelly sizing + 15 checks + sector rotation
| |— seeking_alpha_rss.py     # SA RSS feeds (main + breakfast + dividends + per-ticker)
| |— twelve_data.py           # MACD, Bollinger, EMA, Stochastic, ADX, ATR
| |— yahoo_fundamentals.py     # Fundamentals, earnings calendar, edge scoring
|— .env.template              # API key template (gitignored secrets in Keychain)
|— requirements.txt           # Python dependencies (14 packages)
|— docs/                      # Documentation + PDF guides
```

DeepThinkTrader v3.0 — Execution Intelligence + Self-Improving Learning Loop — Paper Trading Mode Only

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